

Lab 3: Mid-Point Circle Algorithm

LAB SHEET

Theory: Briefly explain Mid-Point Circle algorithm with its algorithm

Lab works:

Question 1: Write a code to draw a circle using the mid point circle algorithm.

```
#include <stdio.h>

#include <graphics.h>

void drawCircle(int x_center, int y_center, int radius) {

    int x = radius;

    int y = 0;

    int err = 0;

    int gd = DETECT, gm;

    initgraph(&gd, &gm, "");

    putpixel(x_center + x, y_center - y, WHITE);

    putpixel(x_center - x, y_center - y, WHITE);

    putpixel(x_center + y, y_center + x, WHITE);

    putpixel(x_center - y, y_center + x, WHITE);

    if (radius > 0) {

        putpixel(x_center - x, y_center + y, WHITE);

        putpixel(x_center + x, y_center + y, WHITE);

        putpixel(x_center - y, y_center - x, WHITE);

        putpixel(x_center + y, y_center - x, WHITE);

    }

}
```

// Initial decision parameter of the midpoint algorithm

err = 1 - radius;

while (x > y) {

 if (err < 0) {

 y++;

 err += 2 * y + 1;

 }

 if (err >= 0) {

 x--;

 err -= 2 * x + 1;

 }

 if (x < y) {

 break;

 }

// Printing the generated point and its reflection in the other octants

putpixel(x_center + x, y_center - y, WHITE);

putpixel(x_center - x, y_center - y, WHITE);

putpixel(x_center + y, y_center + x, WHITE);

putpixel(x_center - y, y_center + x, WHITE);

if (x != y) {

 putpixel(x_center + y, y_center - x, WHITE);

 putpixel(x_center - y, y_center - x, WHITE);

 putpixel(x_center + x, y_center + y, WHITE);

 putpixel(x_center - x, y_center + y, WHITE);

}

```

    }
    delay(5000);
    closegraph();
}

int main() {
    int x_center, y_center, radius;
    printf("Enter the center coordinates (x y): ");
    scanf("%d %d", &x_center, &y_center);
    printf("Enter the radius: ");
    scanf("%d", &radius);
    drawCircle(x_center, y_center, radius);
    return 0;
}

```

Conclusion

- What did u learn?